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Approximate Bayesian Computation with Proper Scoring Rules

An Inference Framework for Calibrated Probabilistic Forecasts

Ashley Turner

Supervisor: Dr Robin Ryder

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Presentation Overview

1. **Introduction:** Research motivation and central question
2. **Background:** Foundational concepts
3. **Methods:** Score-ABC framework and algorithm details
4. **Results:** Empirical performance in MLWP scenario
5. **Discussion:** Contributions and future directions

Central Research Question:
How can proper scoring rules act as principled discrepancy measures in ABC to enable likelihood-free inference of calibrated probabilistic forecasts?

Introduction

Calibration: Real-World Impact

Example: Hurricane evacuation decisions

- Weather model predicts “70% chance of landfall”
- If miscalibrated: actual rate might be 40% or 90%
- Result: unnecessary evacuations or inadequate preparation

Current approach: Augment deterministic models with stochastic components

- Weather/dynamical systems example: perturb initial conditions with arbitrary parameters

Fundamental problem: Difficult to establish a principled method to choose parameters

State of Play: Likelihood-Free Inference

Challenge: Many complex systems have intractable likelihoods

Current approaches and limitations:

- **Traditional ABC:** Uses arbitrary summary statistics (Marin et al., 2012)
- **Distance metrics:** Euclidean, Wasserstein - implicit assumption the model is well-specified

Gap in literature: No principled connection between inference and forecast evaluation

This work's contribution: A framework using proper scoring rules as ABC discrepancies

- Theoretically grounded via scoring rule properties
- Direct alignment with forecast performance criteria, providing insight into impact of parameters on forecast behaviour
- Eliminates arbitrary summary statistic selection

Background

Proper Scoring Rules

Definition: A scoring rule $S(F, x)$ is proper if honest forecasting is optimal:

$$\mathbb{E}_{X\sim\pi}[S(\pi, X)] \leq \mathbb{E}_{X\sim\pi}[S(F, X)] \quad \forall F \in \mathcal{P}(\mathcal{X})$$

Key scoring rules for multivariate forecasts:

- **CRPS (marginal):** $\text{CRPS}(F, x) = \mathbb{E}|X - x| - \frac{1}{2}\mathbb{E}|X - X'|$
- **Energy Score (joint):** $\text{ES}(F, x) = \mathbb{E}\|X - x\| - \frac{1}{2}\mathbb{E}\|X - X'\|$

Both strictly proper; ES reduces to CRPS when $d = 1$. Lower scores represent better performance.

Machine Learning Weather Prediction (MLWP)

Recent advances: Neural networks replace traditional numerical weather prediction

Key advantages of MLWP:

- Orders of magnitude faster than traditional NWP models
- Competitive accuracy for medium-range forecasts

MLWP system used in this report:

- **Model:** U-Net trained on ERA5 reanalysis data
- **Resolution:** 5.625° global grid (64x32 spatial points)
- **Variables:** z500, t850, t2m, u10, v10 + land-sea mask
- **Lead time:** 3-hour forecast

Limitation: Deterministic model has no uncertainty quantification

Uncertainty Quantification via Ensemble Forecasting

Standard approach: Augment base deterministic model with stochastic initial condition perturbations

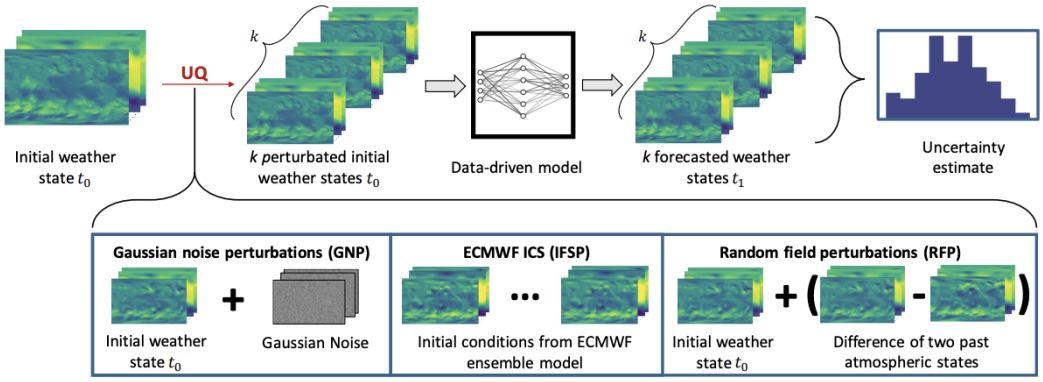
Ensemble generation: $\{f(x_t + e_t^{(k)})\}_{k=1}^{K}$ where $e_t^{(k)}$ are perturbations

Perturbation strategies:

- **Naive Gaussian white noise:** $e_t^{(k)} \sim \mathcal{N}(0, \sigma^2 I)$
- **Spatially correlated noise:** Filtered through covariance kernels, computationally expensive
- **Random field perturbations (RFP):** Sample a "difference field" from historical atmospheric states

Critical question: How to choose perturbation parameters for optimal forecast performance?

Initial Conditions Perturbations for MLWP



Random Field Perturbations (RFP)

Preserves physical balances by sampling differences from historical states

Construction: For an atmospheric variable j, take historical differences
$\Delta z^{(j)}(\tau_1, \tau_2) = z^{(j)}(\tau_1) - z^{(j)}(\tau_2)$

$$e_t^{(k,j)} = \alpha_j \frac{\Delta z^{(j)}(\tau_1^{(k)}, \tau_2^{(k)})}{\|\Delta z^{(j)}(\tau_1^{(k)}, \tau_2^{(k)})\|_E} \tag{1}$$

Advantages:

- Preserves spatial coherence and physical balances by sampling from true historic states
- Seasonally and diurnally matched sampling

Parameter inference problem: Choosing $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_v)$?

Approximate Bayesian Computation (ABC)

Motivation: Inference when likelihood $\mathrm{p}(\mathcal{D}|\theta)$ is intractable

Core idea of ABC: Replace likelihood evaluation with forward simulation

1. Propose parameter $\tilde{\theta}$ from prior $\mathrm{p}(\theta)$
2. Simulate synthetic data $\tilde{\mathcal{D}}$ using $\tilde{\theta}$
3. Compute discrepancy $\rho(\mathcal{D}, \tilde{\mathcal{D}})$
4. Accept $\tilde{\theta}$ if $\rho(\mathcal{D}, \tilde{\mathcal{D}}) \leq \varepsilon$

Challenges in ABC:

- Choice of summary statistics $\mathrm{s}(\mathcal{D})$
- Choice of discrepancy measure $\rho(\cdot, \cdot)$
- Selection of tolerance threshold ε
- Managing parameter dimensionality

Score-ABC Framework

Recap

- Proper scoring rules provide principled forecast evaluation
- RFP preserves physical balances but needs parameter calibration
- Traditional ABC lacks connection to forecast performance

Next: Score-ABC framework

- Theoretical foundation using proper scoring rules as ABC discrepancies
- Algorithmic variants for high-dimensional parameter spaces
- Implementation details for computational feasibility

Score-ABC

Key insight: Replace arbitrary ABC summaries with proper scoring rules

Given forecast-observation pairs $\mathcal{D} = \{(x_t, x_{t+1})\}_{t=0}^{T-1}$:

$$\rho(\mathcal{D}, \tilde{\mathcal{D}}) \to \bar{S}(\pi_{\tilde{\theta}}, \mathcal{D}) = \frac{1}{T} \sum_{t=0}^{T-1} S(\pi_{\tilde{\theta}}(\cdot|x_t), x_{t+1})$$

Advantages:

- Aligns inference with forecast evaluation criteria
- Proper scoring rules enforce honest forecasting
- No arbitrary summary statistics required
- Direct connection to calibration objectives help mitigate the impact of model misspecification

Result: Posterior distribution calibrated to chosen scoring rule

Algorithmic Variants

Challenge: High-dimensional parameter spaces suffer from curse of dimensionality meaning it is difficult to accept proposal parameters

Three algorithmic approaches were tested:

1. Gibbs-ABC: Component-wise parameter updates

- Update parameters α_i sequentially while fixing others
- Mitigates curse of dimensionality

2. SMC-ABC: Sequential Monte Carlo with adaptive tolerance

- Particle-based exploration with resampling
- Adaptive tolerance schedule

3. Optimisation-Style: Gibbs-based score minimisation approach

Gibbs-ABC Algorithm

Update parameters component-wise to avoid curse of dimensionality

Algorithm:

1. **Initialise:** $\boldsymbol{\alpha}^{(0)} \sim \mathsf{p}(\boldsymbol{\alpha})$
2. **For** each iteration $\mathsf{n} = 1, \ldots, \mathsf{N}$:
   - **For** each component $\mathsf{i} = 1, \ldots, \mathsf{v}$:
     - Fix other components: $\boldsymbol{\alpha}_{-\mathsf{i}}^{(\mathsf{n})} = \boldsymbol{\alpha}_{-\mathsf{i}}^{(\mathsf{n}-1)}$
     - Generate M proposals: $\{\alpha_{\mathsf{i}}^{(\mathsf{m})}\}_{\mathsf{m}=1}^{\mathsf{M}}$ from prior
     - Evaluate score for each: $\bar{\mathsf{S}}^{(\mathsf{m})} = \frac{1}{\mathsf{T}} \sum_{\mathsf{t}=0}^{\mathsf{T}-1} \mathsf{S}(\pi_{(\boldsymbol{\alpha}_{-\mathsf{i}}^{(\mathsf{n})}, \alpha_{\mathsf{i}}^{(\mathsf{m})})}(\cdot|\mathsf{x}_{\mathsf{t}}), \mathsf{x}_{\mathsf{t}+1})$
     - Select best: $\alpha_{\mathsf{i}}^{(\mathsf{n})} = \arg\min_{\mathsf{m}} \bar{\mathsf{S}}^{(\mathsf{m})}$
3. **Output:** Parameter trajectory $\{\boldsymbol{\alpha}^{(\mathsf{n})}\}_{\mathsf{n}=1}^{\mathsf{N}}$

SMC-ABC Algorithm

Motivation: Particle-based exploration with adaptive tolerance schedule

Algorithm:

1. **Initialise:** Sample N particles $\{\boldsymbol{\alpha}^{(0,\mathrm{i})}\}_{\mathrm{i}=1}^{\mathrm{N}}$ from prior
2. **For** each generation $\mathrm{t} = 1, \ldots, \mathrm{T}$:
   - Set adaptive tolerance: $\varepsilon_{\mathrm{t}} < \varepsilon_{\mathrm{t}-1}$ (decreasing)
   - **For** each particle $\mathrm{i} = 1, \ldots, \mathrm{N}$:
     - Propose: $\boldsymbol{\alpha}^{(\mathrm{t},\mathrm{i})} \sim \mathrm{K}_{\mathrm{t}}(\cdot|\boldsymbol{\alpha}^{(\mathrm{t}-1,\mathrm{j})})$ with weights
     - Evaluate score: $\bar{\mathrm{S}}^{(\mathrm{t},\mathrm{i})}$
     - Accept if $\bar{\mathrm{S}}^{(\mathrm{t},\mathrm{i})} \leq \varepsilon_{\mathrm{t}}$, otherwise repeat
   - Compute particle weights: $\mathrm{w}^{(\mathrm{t},\mathrm{i})} \propto \mathrm{p}(\boldsymbol{\alpha}^{(\mathrm{t},\mathrm{i})})/\mathrm{q}^{(\mathrm{t},\mathrm{i})}$
   - Resample if effective sample size too low
3. **Output:** Weighted particle approximation of posterior

Implementation

Ensemble generation:

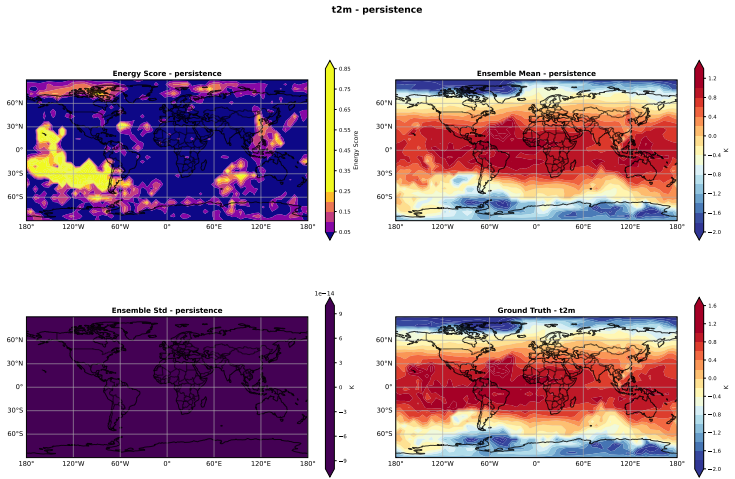
- Ensemble size: K $= 50$ members per evaluation
- GPU vectorisation for parallel RFP sampling
- Temporal resampling: T $= 100$ time steps (unbiased estimator of full history)
- Memory management

Algorithm parameters:

- Proposals per step: M $= 16$ for Gibbs-ABC
- Total iterations: N $= 120$ with burn-in period

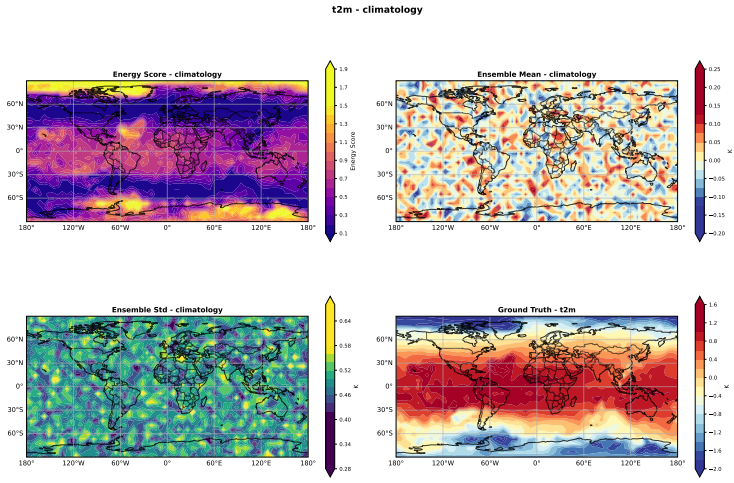
Results

Reference Baseline: Persistence



Persistence forecast: $x_{t+1} = x_t$. Shows fundamental forecast skill benchmark.

Reference Baseline: Climatology



Climatology forecast: long-term seasonal average. Shows fundamental skill floor for weather forecasting.

Reference Baseline: Deterministic + i.i.d. Gaussian

Construction: Add independent Gaussian noise to deterministic MLWP forecasts

$$\mathbf{x}_{t+1}^{(k)} = \mathbf{f}(\mathbf{x}_t) + \epsilon^{(k)}, \quad \epsilon^{(k)} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$$

Choose σ^2 to match empirical forecast residual variance

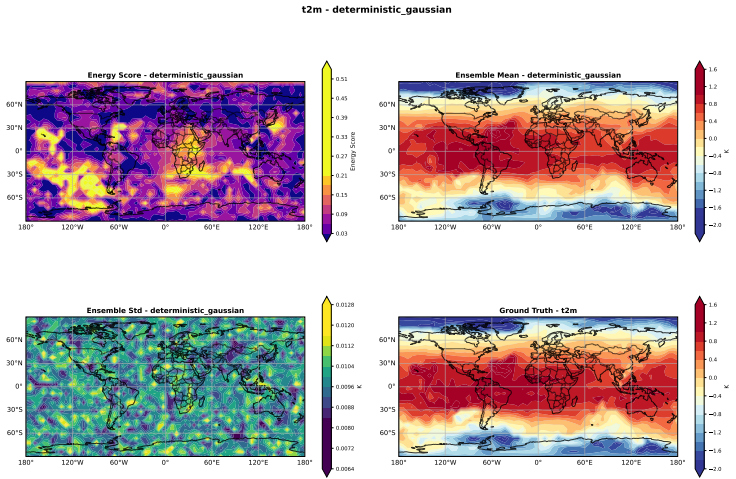
Advantages:

- Marginal calibration is very effective
- Fast ensemble generation

Disadvantages:

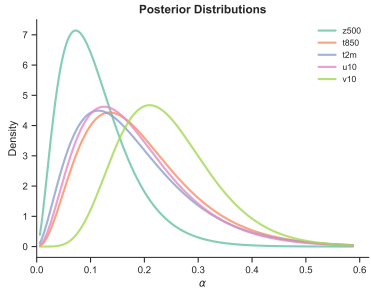
- No spatial correlation structure
- Ignores physical balances
- Unrealistic independence assumptions

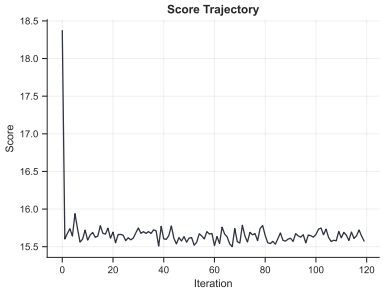
Reference Baseline: Deterministic + i.i.d. Gaussian

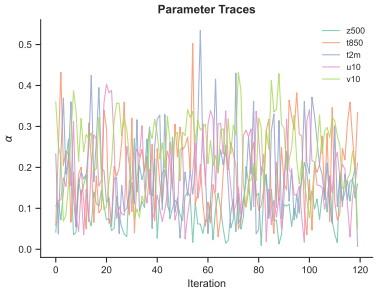


Well-calibrated marginally but lacks spatial coherence—note discontinuous uncertainty patterns.

Gibbs-ABC Results: Energy Score

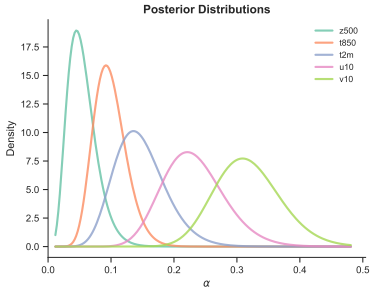


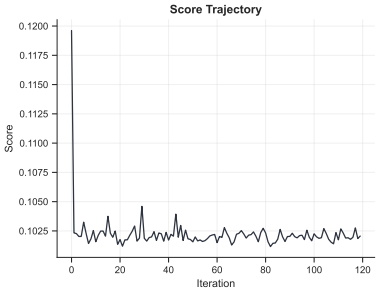


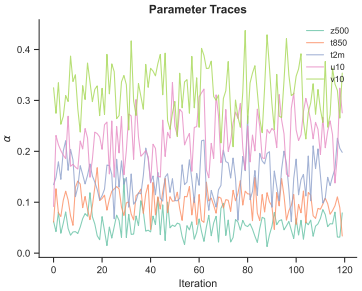


Broad marginal posteriors with early score stabilisation. Energy Score allows parameter compensation.

Gibbs-ABC Results: CRPS

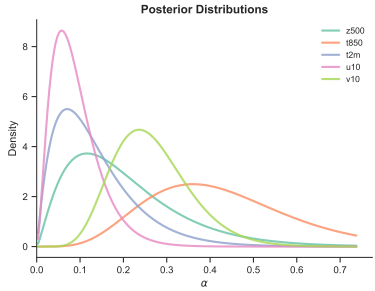


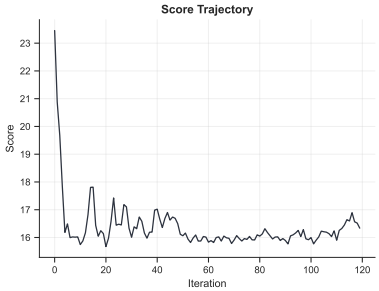


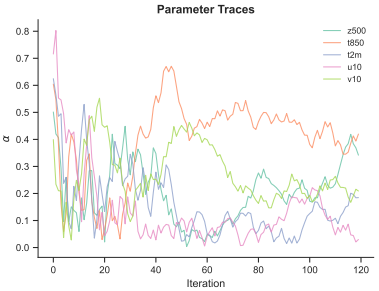


Sharp, ordered marginals with rapid CRPS convergence. Marginal averaging enforces variable-specific calibration.

SMC-ABC Results: Energy Score

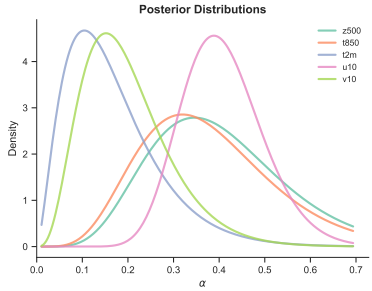


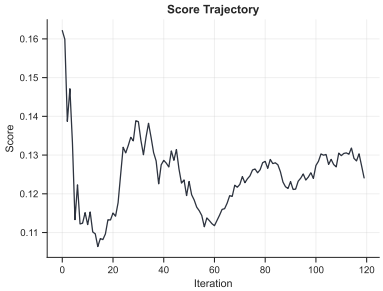


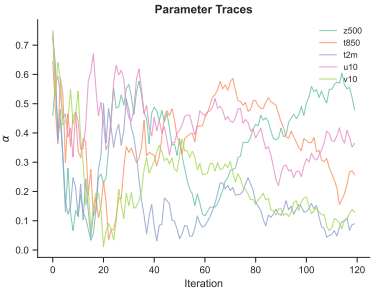


Gradual improvement with adaptive particle evolution. Broader exploration than Gibbs variant.

SMC-ABC Results: CRPS

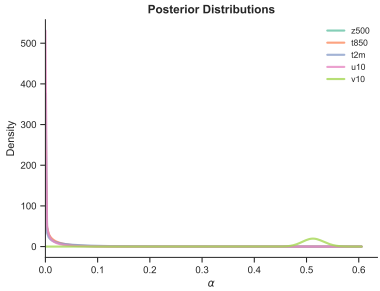


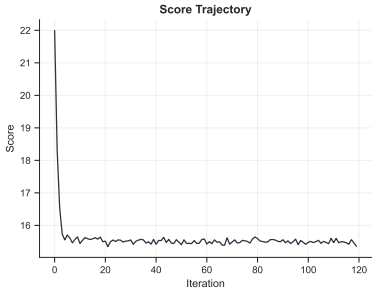


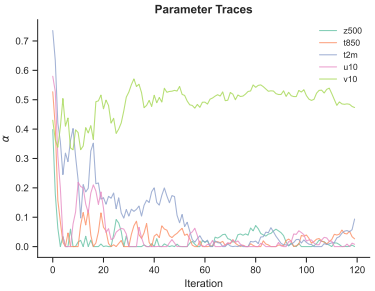


Wide marginals with unstable score trajectory. SMC struggles with CRPS objective.

Optimisation Results: Energy Score

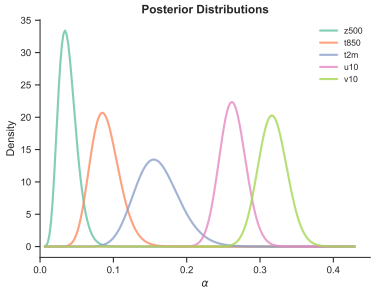


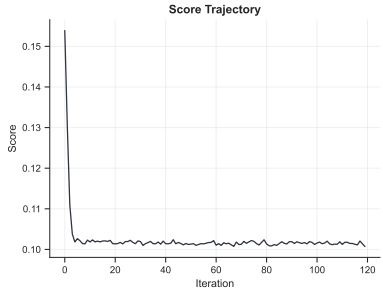


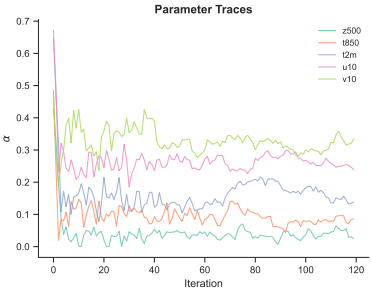


Degenerate distributions indicate local optima. Fast convergence but poor uncertainty representation.

Optimisation Results: CRPS





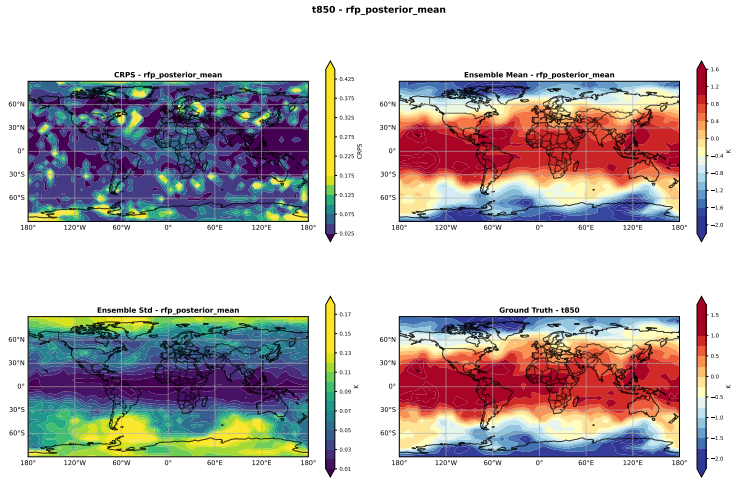


Narrow, ordered posteriors with sharp score improvement. Converges quickly but may miss global optimum.

Performance Comparison: All Methods

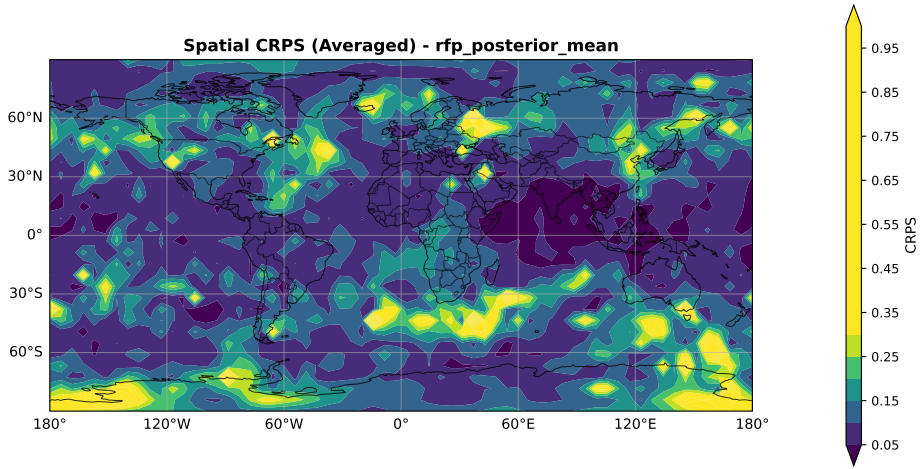
Method	Energy Score	CRPS
Reference Baselines		
Climatology	44.13	0.292
Persistence	30.53	0.156
Deterministic Gaussian	15.49	0.096
RFP Uncalibrated		
Raw RFP ($\alpha = 1.0$)	29.21	0.221
Prior RFP (prior mean)	17.23	0.118
Score-ABC Calibrated		
Gibbs-CRPS	**15.50**	**0.101**
Gibbs-Energy	**15.50**	**0.104**
Optimiser-Energy	**15.43**	**0.105**
Optimiser-CRPS	**15.51**	**0.101**
SMC-Energy	**15.69**	**0.113**
SMC-CRPS	**17.25**	**0.124**

Gibbs-CRPS Sample



Gibbs-CRPS: spatial CRPS, ground truth, ensemble mean, and standard deviation for t850

Spatial CRPS Distribution



Spatially-resolved CRPS shows regionally varying forecast performance with good overall calibration.

Parameter Posteriors Ordering Under CRPS

$$\alpha_{\text{v10}} > \alpha_{\text{u10}} > \alpha_{\text{t2m}} > \alpha_{\text{t850}} > \alpha_{\text{z500}}$$

Physical interpretation:

- **Surface winds (u10, v10):** Highest uncertainty
  - Local topography and friction
  - Boundary layer turbulence
- **Near-surface temperature (t2m):** Moderate uncertainty
- **Upper-level fields (t850, z500):** Lowest uncertainty
  - Large-scale atmospheric processes and smoother dynamics

Discussion

Key Takeaways

Theoretical ideas:

- Proper scoring rules can act as principled ABC discrepancy measures
- Score-informed posteriors can be aligned with forecast evaluation criteria
- Algorithmic variants work in high-dimensional parameter spaces

Practical implications:

- Computationally feasible calibration for MLWP systems
- Framework applicable to many other applications
- Demonstrates the possibilities for simulation-based methods in operational forecasting

Empirical results:

- Competitive performance versus reference methods
- Spatial coherence and physical interpretability maintained

Limitations and Future Directions

Current limitations:

- Computational cost scales with ensemble size and evaluation horizon and is still significant
- Score-ABC may be unsuitable for larger MLWP models
- Temporal resampling may under-represent extreme events

Future research directions:

- Multi-objective Score-ABC combining several scoring rules
- Algorithmic variations
- Extension to other uncertainty quantification methods
- Applications to other fields (epidemiology, finance, climate)

Conclusions

Key findings:

1. Proper scoring rules can serve as principled ABC discrepancy measures
2. Score-ABC enables computationally feasible likelihood-free inference on calibrated parameters
3. The framework maintains spatial coherence while producing competitively-scoring forecasts

Wider implication: Score-ABC provides a framework for likelihood-free calibration of probabilistic forecasting systems, with applications extending to any domain requiring simulation-based uncertainty quantification.

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Thank you.
Questions?